

YANBU PRINCE ABDULAZIZ, Saudi Arabia

WMO: 404390

Lat: 24.144N

Lon: 38.063E

Elev: 26

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.4	55.5	26.3	20.3	77.6	30.6	24.8	76.2	25.5	74.9	23.3	73.9	2.8	30	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.2	111.9	75.6	109.4	75.8	107.3	75.9	86.9	96.9	85.4	95.5	84.2	94.3	17.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
84.5	181.5	92.2	82.8	171.5	90.9	82.1	167.1	90.4	51.9	95.7	50.1	95.1	48.5	95.0	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.9	23.6	21.3	48.6	117.5	2.5	2.6	46.8	119.4	45.3	120.9	43.9	122.4	42.0	124.3		
			41.6	87.7	2.8	2.5	39.6	89.5	37.9	91.0	36.4	92.5	34.3	94.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	70.3	72.5	76.3	82.1	88.5	92.0	92.9	94.1	92.1	87.0	79.7	73.6
	DBStd	9.44	4.56	4.65	4.93	4.82	4.59	4.27	3.66	4.10	3.60	3.53	4.15	4.28
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	9	7	1	0	0	0	0	0	0	0	0	0	1
	CDD50	12218	629	629	814	963	1194	1260	1331	1368	1263	1147	890	730
	CDD65	6753	171	211	349	513	729	810	866	903	813	682	440	266
	CDH74	93990	1474	1946	3773	6507	10653	12653	13719	14511	12366	9257	4737	2394
	CDH80	57732	376	640	1684	3571	6744	8542	9320	10097	8188	5417	2307	846
(14) Wind	WSAvg	8.5	6.9	7.8	8.6	8.5	9.1	10.5	10.3	10.3	9.2	7.5	6.8	6.5
(15) Precipitation	PrecAvg	1.3	0.3	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.3
	PrecMax	4.9	3.1	0.4	3.7	0.4	0.7	0.0	0.0	0.1	0.0	1.0	1.9	2.0
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.3	0.6	0.1	0.7	0.1	0.2	0.0	0.0	0.0	0.0	0.2	0.5	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.6	93.0	98.6	102.6	111.2	114.7	114.6	116.3	112.7	107.2	100.0	93.3
		MCWB	65.1	64.6	67.2	70.0	72.1	73.4	75.0	77.1	75.8	74.3	68.1	66.1
	2%	DB	86.3	89.3	94.0	98.9	107.4	111.0	109.8	111.4	109.1	102.5	95.7	89.7
		MCWB	65.8	65.2	67.5	71.0	72.6	74.4	76.4	77.9	76.5	72.9	69.8	66.7
	5%	DB	84.2	86.4	91.3	96.8	103.8	107.8	107.3	107.8	105.9	100.2	93.2	87.5
		MCWB	65.9	66.3	68.2	71.1	73.4	74.9	76.5	77.8	76.7	75.0	70.6	67.2
	10%	DB	82.0	84.1	88.2	94.2	100.5	104.7	104.0	104.4	102.6	97.5	91.1	84.6
		MCWB	65.1	66.1	67.3	70.3	72.7	74.7	76.0	77.6	77.2	76.1	70.9	66.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.5	75.5	77.0	79.9	83.3	85.3	87.2	88.7	87.9	87.0	81.7	77.1
		MCDB	82.1	83.7	85.9	90.1	94.3	97.9	98.1	99.0	98.5	96.0	90.0	84.3
	2%	WB	72.5	73.0	74.7	77.4	80.9	83.1	85.4	87.2	85.6	84.3	78.5	74.0
		MCDB	80.5	81.6	85.2	88.8	92.6	95.6	96.1	96.8	95.4	92.8	87.6	82.7
	5%	WB	70.0	71.2	72.8	75.6	79.1	81.4	83.8	85.9	84.1	82.1	76.3	71.8
		MCDB	79.6	80.8	84.6	88.1	91.8	93.6	94.1	95.5	93.9	90.9	86.0	81.6
	10%	WB	67.5	69.3	70.7	73.9	77.5	80.1	82.2	84.4	82.6	80.3	74.4	69.5
		MCDB	78.0	79.9	83.2	87.7	91.0	92.8	92.4	94.6	92.9	89.6	84.7	80.2
(35) Mean Daily Temperature Range	5% DB	MDBR	22.9	23.4	24.5	24.4	24.8	26.0	23.3	22.2	23.5	22.8	23.3	23.5
		MCDBR	24.0	25.0	26.3	26.0	27.9	29.3	27.5	26.4	27.0	26.0	25.4	25.9
		MCWBR	13.1	13.1	13.0	12.5	12.9	14.1	13.9	13.9	13.5	14.0	12.4	12.9
	5% WB	MCDBR	19.2	19.1	21.8	22.3	24.2	25.7	22.8	19.7	21.4	21.7	21.5	20.5
		MCWBR	13.0	12.7	13.3	13.1	12.9	13.5	13.0	12.4	12.2	13.0	11.7	11.9
(40) Clear-Sky Solar Irradiance	taub		0.390	0.449	0.480	0.517	0.534	0.522	0.550	0.592	0.542	0.500	0.435	0.410
	taud		2.280	2.077	1.980	1.915	1.877	1.916	1.888	1.807	1.888	1.991	2.162	2.210
	Ebn at Noon		273	263	261	254	248	249	242	232	241	244	255	260
	Edh at Noon		38	50	57	62	65	62	63	69	62	53	43	39
(44) All-Sky Solar Radiation	RadAvg		1434	1713	2078	2312	2439	2551	2453	2284	2107	1814	1508	1348
	RadStd		73	68	79	71	81	37	49	56	55	45	48	35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.90	N/A	N/A	N/A	N/A	N/A	N/A	+302	+317
(47) Regional (1 neighbor)		+0.83	N/A	-1.55	N/A	N/A	N/A	N/A	+281	+275

Nomenclature: See separate page